X/K/TY

Reg. No: 23BCE0444

VIT
Vellore Institute of Technology

Final Assessment Test – November 2025

Course: BCSE303L - Operating Systems
Class NBR(s): 1314/1317/1319/1321/1323/1327/1331/1334/1337/1339/1341/1342/1346/1350/1354/1358/1362/1365/2375/4523
Slot: C2+TC2
Time: Three Hours
Max. Marks: 100

- KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE
- DON'T WRITE ANYTHING ON THE QUESTION PAPER

COs	CO Statements
CO1	Interpret the evolution of OS functionality, structures, layers and apply various types of system calls of various process states
CO2	Design scheduling algorithms to compute and compare various scheduling criteria.
CO3	Apply and analyze communication between inter process and synchronization techniques.
CO4	Implement page replacement algorithms, memory management problems and segmentation.
CO5	Differentiate the file systems for applying different allocation, access technique, representing virtualization and providing protection and security to OS.

BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)

Answer ALL Questions
(10 X 10 = 100 Marks)

1. Demonstrate, how the design of an operating system for general-purpose systems balance functionality, security, and performance, and discuss the trade-offs involved in achieving this balance? **CO1 BL2**

2. i. Compare and analyze a thread and a process by applying the following criteria:- **[5] CO1 BL3**
   - Complexity
   - Structure
   - Switching overhead
   - Benefits

   ii. Explain the purpose of system calls in an operating system and describe the purpose and identify the type (privileged or non-privileged) of each of the following system calls:- **[5]**
   ptrace() setuid() execve(), read(), init_module()

3. Consider the following set of processes that arrive at different times, with their respective burst times. Using the Shortest Job First (SJF) and Round Robin (quantum-4sec) scheduling algorithms. Illustrate a complete Gantt chart and calculate the Completion Time (CT), Average Turnaround Time (TAT), and Average Waiting Time (WT) for each process. CO2 B

Process	Arrival Time(AT)	Burst Time(BT)
P1	0	8
P2	1	4
P3	2	9
P4	3	5

4. Analyze the information in the table and draw the resource allocation graph (RAG). Apply the Banker's algorithm to determine, whether the system as described in the table will be in a safe state or not. CO2

Process	Allocated				Maximum Demand				Available			
	R1	R2	R3	R4	R1	R2	R3	R4	R1	R2	R3	R4
P0	2	0	0	1	4	3	2	2	4	3	2	2
P1	1	1	0	0	1	2	1	3				
P2	1	0	1	0	4	3	4	3				
P3	0	1	0	1	3	5	3	2				
P4	2	2	1	1	3	4	3	4				

5. Explain how semaphores are used to manage concurrency in operating systems, with a focus on synchronization problems such as the Producer-Consumer scenario. Compare binary and counting semaphores, and justify which type is more suitable for solving this problem. Provide a pseudocode solution to demonstrate your explanation. CO3

6. Narrate the needs for critical region and provide relevant justifications. Give an example of a scenario where a binary semaphore would be more appropriate than a lock. CO

Demonstrate how Peterson's Algorithm resolves process synchronization issues for any real-world problem.

7. Compare contiguous, linked and indexed file allocation in terms of space utilization, access time and fragmentation. CO5 BL4

For a system handling large, frequently updated files with both sequential and random access. Discuss which method is best and why? Support your choice with technical reasoning.

8. Evaluate how the following parameters influence the overall hard disk throughput. CO5 BL5

- Seek Time
- Rotational Delay
- Actual Transfer Time

Demonstrate the working of Shortest Seek Time First (SSTF) , C-LOOK, and SCAN algorithm for the given specifications:-

- Request sequence = 82, 170, 43, 140, 24, 16, 190
- Initial head position = 50
- Disk size : 0-199

9.a) Compare Paging, Segmentation and Paged Segmentation in terms of memory allocation, address translation and fragmentation. CO4

Then, for a system with 64 MB physical memory, a 32-bit virtual address space, and a 4 KB page size, calculate the page table size and explain how its size (smaller or larger than a page) affects memory management and the choice between the three schemes.

OR

9.b) A computer system uses paging for memory management, with system details:- CO4

Logical address space: 32 KB, Page size: 2 KB, Physical memory size: 16 KB and Each frame is 2 KB (same as page size).

The page table for a process is:

Page Number	Frame number
0	3
1	0
2	2
3	1

- Calculate the number of bits in Logical address, Offset, Page number.
- Convert the following logical addresses 8165, 945, 256 into physical address.
- Will the logical address 16384 cause a page fault? Why or why not?

Consider three users: X, Y and Z and three files: file1.txt, file2.sh and file3.cfg. The access permissions (Access Control List - ACL) for each user on each file are given below: CO5 BL1

File	User	Read(R)	Write(W)	Execute(EXE)
File1.txt	X	1	1	0
	Y	1	0	0
File2.sh	X	0	0	1
	Z	1	1	1
File3.cfg	Y	1	1	0
	Z	1	0	0

- Construct the Capability List for each user based on the above ACL data.
- Identify which users have both read and write access to file3.cfg.
- Determine whether X has execute permission for file2.sh. Justify your answer.

OR

w does virtualization improve resource utilization and scalability in modern nfrastructures, and what are the potential drawbacks or challenges ciated with its implementation? CO5

⇔⇔⇔ X/K/TY ⇔⇔⇔